Solution:

$$f(x) = \int_{-\infty}^{\infty} f(x, y)dy$$

$$f(x) = \frac{1}{10}\int_{1}^{2}(4x-2y)dy$$

$$= \frac{1}{10}\left[4xy - \frac{2y^2}{2}\right]_1^2$$

$$= \frac{1}{10}[4x(2-1)-(4-1)] = \frac{1}{10}(4x - 3)$$

$$f(y) = \int_{-\infty}^{\infty} f(x, y)dx$$

$$f(y) = \frac{1}{10}\int_{1}^{3}(4x-2y)dx$$

$$= \frac{1}{10}\left(\frac{4x^2}{2} - 2xy\right)_1^3$$

$$= \frac{1}{10}\left(2x^2 - 2xy\right)_1^3$$

$$= \frac{1}{10}\left[2(9-1)-2y(3-1)\right]$$

$$= \frac{1}{10}(2\times 8 - 6y + 2y)$$

$$= \frac{1}{10}(16 - 4y) = \frac{1}{5}(8 - 2y)$$

Marginal density function of X

$$f(x) = \begin{cases} \frac{1}{10}(4x-3) & , 1 \le x \le 3 \\ 0 & , \textit{elsewhere} \end{cases}$$

Marginal density function of Y

$$f(x, y) = \begin{cases} \frac{1}{5}(8-2y) & , 1 \le y \le 2 \\ 0 & , \textit{elsewhere} \end{cases}$$

9.6 Mathematical expectation

Probability distribution gives us an idea about the likely value of a random variable and the probability of the various events related to random variable. Even though it is necessary for us to explain probabilities using central tendencies, dispersion, symmetry and kurtosis. These are called descriptive measures and summary measures. Like frequency